Max Chan

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EDUCATION

Nanyang Technological University, Singapore Aug 2024 – May 2028
Bachelor of Computing in Computer Science

Tampines Meridian Junior College, Singapore Jan 2020 – Dec 2021
Singapore-Cambridge GCE A-Levels

PROJECTS

Human vs Robot Box-Stacking Game github.com/ka1dus/so101_stack_game
Python, OpenCV, MuJoCo, LeRobot, PyTorch

- Built a turn-based game where a person and an SO-101 arm take turns stacking boxes, driven by a state machine (setup, human turn, robot turn, settle, judge) that runs unchanged on the real arm, in MuJoCo simulation or on recorded video for testing.
- Wrote a camera-based referee that judges every turn, working out which box was taken, whether it landed on the stack and whether anything fell, and asks the player on screen only when unsure. It called 75 of 75 turns correctly with zero false losses across 10 simulated games.
- Built a safety layer for working next to people, with a hand-detection thread that freezes the arm when a hand enters the workspace and resumes 1 s after it leaves, an e-stop that cuts motor torque, and a watchdog so the game never hangs.
- Ran full games against real players and fixed what broke, including early key presses ending turns, the robot misjudging which box it held and false calls when a box nudged the stack, replaying every recorded turn to confirm each fix changed nothing else.
- Trained the robot's place skill on 7,000 synthetic demonstrations in a MuJoCo digital twin, and it placed 8 of 10 boxes in real games.

WORK EXPERIENCE

Botmind July 2026 – Dec 2026
Robot Engineering Intern

- **Real-Time Robot Surveillance Perception** (PyTorch, TensorRT, CUDA). Ran six detectors concurrently on one GPU feed, combining YOLO, ByteTrack, pose estimation and monocular depth in a threaded runtime, then exported to TensorRT FP16 for a 1.8–2.4× speedup (up to 365 FPS) with over 99.6% recall parity against PyTorch.
- Integrated detections into the robot fleet’s REST API, so each event raises an alert tied to its robot and camera, sent from a background queue with OAuth auto-renewal.
- Built Grafana dashboards over robot fault-reporting logs to pinpoint errors in deployed robots.
- Field-deployed real-time PPE detection on a quadruped robot running on an edge device, from concept to a live client demo in an aircraft hangar in under 3 days.

JIMAICONSULTING PTE LTD Nov 2025 – Present
Co-Founder

- Built an LLM agent that qualifies leads over WhatsApp and books appointments through tool calls to Google Calendar, with idempotent triggers and retries in production.
- Co-founded the company, reaching ~$10,000 revenue in the first 2 months and 90% client retention after month one.

SKILLS

Robotics: LeRobot, imitation learning (ACT), MuJoCo, inverse kinematics, domain randomization, digital twins, teleoperation
Vision: PyTorch, YOLO, OpenCV, TensorRT, CUDA, ONNX, ByteTrack, pose estimation, camera calibration
Programming: Python, C++, SQL
Systems: Linux (WSL), Git, REST APIs
Languages: English, Chinese